

Lower time to market

The SoM approach allows the fastest time to market. Utilising off-the-shelf standard components like a baseboard and SoM, designers are able to start developing applications even before the custom baseboard is complete. As mentioned earlier, this allows software development and hardware development to proceed simultaneously, greatly reducing the time to get the product to the market.

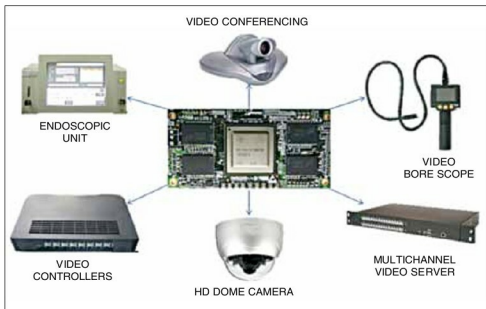
Working with an SoM

An SoM designer needs to be an expert in different aspects of product design in order to conceptualise and evaluate ideas, and make them tangible through products. For designing and supporting a high-quality SoM, he needs to have:

1. Deep knowledge of SoC, memory, power management, etc
2. Knowledge of software infrastructure and firmware
3. Understanding of system design to define correct SoM and connectivity, for development of different products

Fitting a processor, memory, flash and interfaces within a small space as in a typical chip, requires careful layout with respect to environmental conditions such as temperature. Simulations are followed by extensive validation testing to ensure that the SoM solution, including a heat-sink, meets its specifications. In addition, rigorous testing is done to ensure that the clock, power, reset and memory signals have sufficient margin to guarantee consistent operation over the lifetime of the product.

As SoM-based designs are similar to microcontroller-based board



System-on-module applications

designs, it will be easier for microcontroller-based systems to transit to SoMs.

What next?

The semiconductor industry is witnessing blazingly fast changes. Different companies use different strategies to improve their portfolio.

"Current SoM-based design requires the use of connectors. To reduce BOM cost, we are working on connector-less SoMs that can be directly soldered onto customer boards. This will make SoMs usable for very high-volume applications too. We are also working on bringing in high-performance microprocessor based SoMs to India. One example is OMAP4460/70, which is a dual-core Cortex A9 based processor," says Jithu Niruthambath, joint director at Calixto Systems.

As regards market trends in this field, Jithu shares, "We are seeing customers who have started using SoMs

for very high-volume applications such as tablets as well. This shows that SoM-based solutions are viable for low- and high-volume applications equally."

Nigel Forester, market development manager at Embedded Computing of Emerson Network Power, expects to see a much broader portfolio of devices being used on CoMs in the future, which will broaden the use cases.

"Historically, CoMs have typically been based on Intel x86 architecture devices, but last year Emerson announced a range of CoMs based on the Freescale QorIQ range of processors. As compute-density and performance increase, more customers are adopting plug-on compute modules. This, in turn, is driving a greater range of CoMs that are more application-specific," Forester adds. ●

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